



UMD DATA605 - Big Data Systems

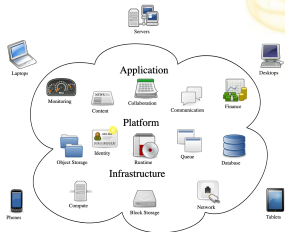
11.1: Cloud Computing

- **Instructor:** Dr. GP Saggese, gsaggese@umd.edu

- *Cloud Computing*

Cloud Computing

- Computing as **"service" rather than "product"**
 - Storage and computing in the cloud
 - Edge devices (e.g., phones, laptops, tablets) interact with the cloud



- **Advantages of cloud computing**
 - Device agnostic: seamless computation across devices
 - On demand
 - Efficiency / scalability
 - Programming frameworks (e.g., Hadoop, Spark, Dask) enable scalability
 - Reliability
 - Cost: “pay-as-you-go” for resources
 - Cheaper than building own systems
 - Computing as a commodity (like electricity)

Buying infrastructure

- **To buy or to rent?**
- **Building / buying infrastructure**
 - Require time and capital investments (Capex)
 - Especially at the beginning without revenues
 - Smooth cash flow (constant \$/mo) is better than lumpy one (big one-time purchase)
 - Buy hardware (e.g., computers, storage, network)
 - Estimate current hardware size
 - Difficult to estimate future demands
 - Update obsolete hardware
 - Cost of owning hardware (Opex)
 - Data center, electricity, cooling, handling faults
 - Administering
 - Install, update, maintain software stack

Renting infrastructure

- **Renting infrastructure** (i.e., cloud computing)
 - Pay for what you use
 - Low initial capital investment
 - Ready systems with a click
 - No multi-year resource plan needed
 - Choose machines for your application and data needs

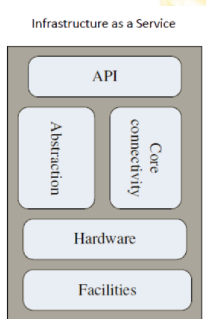
Cloud Computing

- **Ideas of cloud computing around for a long time**
 - Mainframes + thin clients (1960s)
 - Personal computers (1980s)
 - Grid computing for supercomputers (1990s)
 - Peer-to-peer architecture (early 2000s)
 - Client-server model (Web 1.0 and Web 2.0)
 - Cloud computing (2010s)
- **Now, it finally works**
- **Why now?**
 - A convergence of key technologies
 - OS virtualization
 - Large data centers
 - Decreasing hardware costs
 - Big data frameworks



Infrastructure-as-a-Service (IaaS)

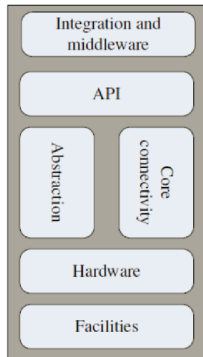
- Cloud provides low-level resources like virtual machines, storage, networking
 - Users install and maintain the operating system and applications
- **Examples**
 - AWS EC2
 - Google Compute Engine
 - Microsoft Azure Virtual Machines
 - IBM Cloud Infrastructure
- **Benefits**
 - Flexibility in managing resources
 - Scalability to meet demand
 - Cost-effective, pay for what you use
- **Use Cases**
 - Hosting websites and applications
 - Data analysis and processing
 - Development and testing environments



Platform-as-a-Service

- **Problem:** assembling your own software stack requires work
 - Install
 - Configure
 - Manage dependencies (e.g., incompatible versions)
- **Solution:** get a pre-built software stack
 - Software stack as a virtualization solution (e.g., VMware or Docker)
 - Pre-installed OS
 - Libraries
 - Application software
- **Business model built around this**
 - E.g.,
 - Google App Engine
 - Managed Hadoop
 - E.g., pre-built images for Hadoop
 - Hortonworks, Cloudera
 - E.g., pre-built distributions for Linux
 - RedHat, Gentoo, CentOS

Platform as a Service



Software-as-a-Service (SaaS)

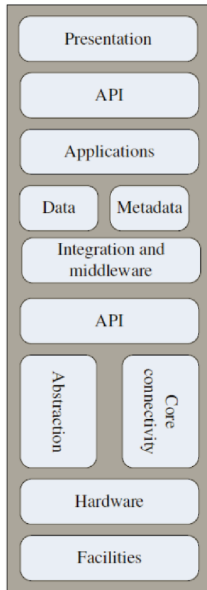
- **Cloud provides the application**

- No need to install on your machine
- Use directly from the cloud
- Examples include:
 - *Dropbox*: Access and share files without local storage
 - *Salesforce*: Manage customer relationships online
 - Any app running in a browser: Google Docs or Microsoft Office 365

- **Benefits**

- Accessibility from any device with internet connectivity
- Automatic updates and maintenance by the provider
- Scalability for growing user needs
- Cost-effectiveness by reducing physical hardware and software installations

Software as a Service



X-as-a-Service

- After 2010, business model of **X-as-a-service (XaaS)**
 - Mobility-as-a-service (e.g., Uber)
 - Games-as-a-service (e.g., Google Stadia)
 - Storage-as-a-service (e.g., S3, Google Drive)
 - Desktop-as-a-service (e.g., AWS app)
 - Marketing-as-a-service
 - Banking-as-a-service
 - ...

Cloud Deployment Models

- **Private**

- Internal cloud hosted on organizational premises
- E.g., company's data center running virtualized services

- **Public**

- External cloud hosted by third-party providers
- E.g., AWS, Azure, GCP offering compute and storage to clients

- **Hybrid cloud**

- Combine private and public environments to optimize cost, security, scalability
- Sensitive workloads stay internal
- Scalable tasks move to public cloud

- **On-premises resources**

- E.g., corporate servers in a building

- **Off-premises resources**

- E.g., cloud provider's distributed data centers

